

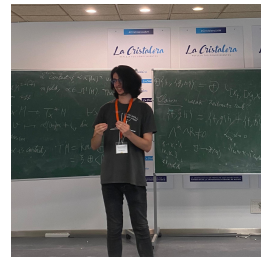
Rubén Izquierdo-López, PhD student

✉ ruben.izquierdo@icmat.es

📞 On request

🌐 My personal webpage: rubenizquierdolopez.xyz

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Teaching experience

- | | |
|------------------------------|---|
| December 2025-now | 📖 Teaching assistant at UAM (Universidad Autónoma de Madrid) teaching |
| | <ul style="list-style-type: none">• Linear Geometry (2025–2026) for second year students of Mathematics and Computer Science.• Linear Geometry (2025–2026) for first year students of Mathematics. |
| September 2024-December 2025 | 📖 Professor at UNIR (Universidad Internacional de la Rioja) teaching |
| | <ul style="list-style-type: none">• Differential Equations (2025–2026) for 2nd year students of Physics.• Advanced Algorithms (2025–2026) as an optative course for students of Computer Science.• Advanced Algorithms (2025–2026) for 3rd year students of Computational Mathematics.• Dynamical Systems (2024–2025) for 4th year students of Computational Mathematics• Computational Logic (2024–2025) for 1st year students of Computational Mathematics.• Complex Variables (2024–2025) for 3rd year students of Physics.• Numerical Analysis (2024–2025) for 2nd year students of Physics.• Differential Equations (2024–2025) for 2nd year students of Physics. |
| 2022-2023 | 📖 Teaching assistant in “Matemáticas Básicas” at Universidad Complutense de Madrid. |
| 2020-2022 | 📖 Support Teacher at Savannah SLC in Torrejón de Ardoz. |

Research experience

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|-------------------|--|
| December 2025-now | 📖 FPU2024 fellowship for developing my PhD in CSIC (Instituto de Ciencias Matemáticas). |
| 2023-2024 | 📖 Severo Ochoa contract for Master students in CSIC (Instituto de Ciencias Matemáticas) supervised by Prof. Manuel de León. |
| 2023 | 📖 JAE Intro research grant supervised by Prof. Manuel de León. |

Education

- 2024– now I am currently a PhD student in Mathematics under the supervision of prof. Manuel de León in Universidad Autónoma de Madrid.
- 2023–2024 **Master's degree in Advanced Mathematics, (average of 9.77/10)** in Universidad Complutense de Madrid.
Thesis title: *Coisotropic reduction in Multisymplectic Geometry*. Advisor: *Manuel de León Rodríguez*. **Grade:** 10.
- 2019–2023 **Bachelor's degree in Mathematics, (average of 9.44/10)** in Universidad Complutense de Madrid. I was awarded with "premio excelencia UCM", for one of the best academic records in Mathematics.
Thesis title: *Simplicial and Cellular Homology*. **Advisor:** *Manuel Alonso Morón*. **Grade:** 9.7/10.

Languages

- Spanish: Native
- English: C1, 100/120 on TOEFL.

Grants

- Severo Ochoa grant for Master students in academic year 2023-2024.
- "Excellence grant" of Comunidad de Madrid in academic years 2020-2021, 2021-2022, 2022-2023.
- JAЕ Intro research grant.

Schools and conferences

- March 2026 **XX International Young Researchers Workshop on Geometry, Dynamics and Field Theory** in Loughborough University, United Kingdom (presented a poster on "Higher form symmetries and general conservation laws").
- January 2026 **VIII Iberoamerican meeting on Geometry, Dynamics and Field Theory** in Universidad de la Laguna, Tenerife (talk on "The graded Poisson bracket of general conservation laws in classical field theories").
- September 2025 **XXXIII International Fall Workshop on Geometry and Physics** in Universidad de Córdoba (Plenary talk on "Graded brackets in classical field theories: Conservative and dissipative").
- June 2025 **Summer School GDFT 2025** in Miraflores de la Sierra, Madrid (talk on "Extensions of brackets in classical field theories").
- January 2025 **Geometry and Topology for the Future III** in Universidad Complutense de Madrid, España (talk on "Recent Developments in the Theory of Brackets in Classical Field Theories").
- January 2025 **XIX Young Researchers Workshop in Geometry Dynamics and Field Theory** in University of Verona, Verona, Italy (talk on "Graded structures of classical field theories").
- November 2024 **VII Congreso de Jóvenes Investigadores de la RSME** in Universidad del País Vasco, Bilbao, Spain (talk on "Graded brackets in classical field theories").
- November 2024 **New Advances in Contact and Symplectic Hamiltonian Dynamics** in Great Bay University, Dongguan, Guangdong, China (talk on "Poisson and Dirac structures in Classical Field Theories").

Schools and conferences (continued)

- September 2024  **Workshop de Jóvenes Investigadores** in UCM, Madrid (talk on "Coisotropic reduction in Classical Field Theories").
- August 2024  **Workshop on Geometric Aspects of Material Modelling** in UNED, Madrid (talk on "The Multisymplectic framework of Field Theories").
- June 2024  **XVI International ICMAT Summer School on Geometry, Dynamics and Field theory** in Miraflores de la Sierra, Madrid.
- February 2024  **The Geometry of Field Theories** in Real Colegio de Santa Cristina, Madrid (short talk on "Coisotropic reduction in Multisymplectic Geometry").
- February 2024  **XVIII International Young researchers Workshop in Geometry, Dynamics and Field Theory** in Warsaw University (short talk on "Coisotropic reduction in different phase spaces").
- January 2024  **XXV Winter Meeting on Geometry, Dynamics and Field Theory** in Universidad de Zaragoza.
- December 2023  **de León fest** in ICMAT.
- November 2023  **BYMAT** in ICMAT (short talk on "Coisotropic reduction in symplectic, cosymplectic, contact and cocontact geometry").
- July 2023  **JAЕ summer School** in ICMAT.

Talks and seminars

- May 2025  **Coloquio Junior de matemáticas**, ICMAT, Madrid (talk on "Natural operations in differential geometry").
- February 2025  **Q-math seminar**, Universidad Carlos III de Madrid (talk on "Graded Poisson brackets in classical field theories").
- September 2024  **Seminario de doctorandos** in UCM, Madrid (talk on "Geometría multisimpléctica, el marco para teoría clásica de campos").
- June 2024  **Gamma Seminar**, online (talk on "Coisotropic reduction in Multisymplectic Geometry").

Volunteering

- July 2024  **Assistant in ECM (European Congress of Mathematics)** at Universidad de Sevilla.
- April 2023  **Assistant in "Concurso de Primavera"** at Universidad Complutense de Madrid.

Research Publications

Journal Articles

- 1 M. de León, R. Izquierdo-López, and X. Rivas, "Brackets in Multicontact Geometry and Multisymplectization," *Mediterr. J. Math.*, vol. **23**, no. 2, p. 81, 2026, 10.1007/s00009-026-03077-4.
- 2 M. de León and R. Izquierdo-López, "Graded poisson and graded dirac structures," *Journal of Mathematical Physics*, vol. 66, no. 2, p. 022 901, 2025, ISSN: 0022-2488.  DOI: 10.1063/5.0243128.
- 3 M. de León and R. Izquierdo-López, "A review on coisotropic reduction in symplectic, cosymplectic, contact and co-contact hamiltonian systems," *Journal of Physics A: Mathematical and Theoretical*, vol. 57, no. 16, p. 163 001, 2024.  DOI: 10.1088/1751-8121/ad37b2.
- 4 M. de León and R. Izquierdo-López, "Coisotropic reduction in multisymplectic geometry," *Geometric Mechanics*, vol. 01, no. 03, pp. 209–244, 2024.  DOI: 10.1142/S2972458924500096.

Preprints

- 1 M. de León, R. Izquierdo-López, and M. Láinz, *The multisymplectic group*, In preparation, 2026.
- 2 M. de León, R. Izquierdo-López, J. G. Rifa, and M. D. Maceda, *Relationship between the observables in the space of solutions and the multisymplectic framework*, In preparation, 2026.
- 3 M. de León, R. Izquierdo-López, X. Rivas, and F. SandersI, *The split spacetime formalism in multicontact field theories*, In preparation, 2026.
- 4 M. D. León, R. Izquierdo-López, L. Schiavone, and P. Soto, *Regularization of singular time-dependent lagrangian systems*, 2026. arXiv: 2603.24308 [math.DG].  URL: <https://arxiv.org/abs/2603.24308>.
- 5 R. Izquierdo-López, M. de León, L. Schiavone, and P. Soto, *A zoo of coisotropic embeddings*, 2025. arXiv: 2509.19039 [math.DG].  URL: <https://arxiv.org/abs/2509.19039>.
- 6 M. de León and R. Izquierdo-López, *A description of classical field equations using extensions of graded poisson brackets*, 2025. arXiv: 2507.04743 [math-ph].  URL: <https://arxiv.org/abs/2507.04743>.

Peer review

-  Journal of Nonlinear Science
-  Mediterranean Journal of Mathematics
-  Geometric Mechanics

Links

-  **Google Scholar:** <https://scholar.google.es/citations?user=vQKgf6sAAAAJ&hl=es>. **Citations:** 30
-  **Mathscinet:** <https://mathscinet.ams.org/mathscinet/author?authorId=1609833>. **Citations:** 4 citations by 11 different authors.
-  **Scopus:** <https://www.scopus.com/authid/detail.uri?authorId=58553020100>. **Citations:** 5